PRODUCT DEMO · HINGLISH WALKTHROUGH

AI Workforce

Health-plan aur PBM member services ke liye AI voice CSR agents

Ek AI Customer Service Representative (CSR) jo phone pe member calls uthata hai, coverage, benefits, claims, prior-authorization, aur pharmacy-network se juda sawaal khud hi resolve karta hai — aur jab nahi kar paata, tabhi human ko handoff karta hai. Yeh document har screen, use case, tareeka, aur cost/resolution ka math — sab kuch Hinglish mein samjhata hai.

BETA v0.1 — active pilot build, abhi production-certified nahi hai

Taiyaar kiya gaya: 26 Aug 2026
Internal product walkthrough

01 — EXECUTIVE SUMMARY

Yeh banaya hi kyun gaya

MEMBER SERVICES MEIN PROBLEM KYA HAI

- Repetitive, structured sawaalon pe call volume bahut zyada hai: coverage check, prior authorization (PA) status, in-network pharmacy dhoondhna, claim status, benefits samjhana.
- Ek human-handled contact-center call ka fully-loaded cost roughly $10–15 padta hai, aur headcount badhane se cost linearly badhta hai.
- Quality analyst-to-analyst alag hoti hai; QA review manual hoti hai aur sirf kuch % calls hi sample ho paati hain.
- Business hours ke baad coverage nahi hoti, aur peak time pe queue lambi ho jaati hai.

AI WORKFORCE ISKO KAISE SOLVE KARTA HAI

- LLM voice agent jo live PBM data (eligibility, claims, formulary, pharmacy network) ke saath function/tool-calling karta hai — yeh scripted IVR tree nahi hai.
- Aapke apne uploaded plan documents pe Retrieval-Augmented Generation (RAG), citations ke saath — matlab jawab grounded hote hain, hallucinated nahi.
- Supervisor cockpit (Live Operations) — real-time confidence score aur ek-click human takeover, matlab kabhi bhi kuch fully unsupervised nahi chalta.
- Har call ko waise hi score kiya jaata hai jaise QA analyst karta (accuracy, compliance, empathy, resolution) — automatically, sample pe nahi, 100% calls pe.

IS DOCUMENT MEIN KYA HAI

Section 2 poori technical architecture end-to-end samjhata hai. Section 3 har screen ka walkthrough deta hai — Dashboard, Agent Studio, Live Operations, Conversations, Customers, Workflows, Analytics, AI Training Center, Data Import — har ek ka use case, kaam karne ka tareeka, business value, aur har button ki table. Section 4 un 12 actions ("tools") ko list karta hai jo AI call ke dauraan kar sakta hai. Section 5 dashboard ke cost aur resolution wale numbers ka formula-by-formula matlab samjhata hai. Section 6 data handling cover karta hai. Section 7 honest beta disclaimer aur roadmap hai.

02 — ARCHITECTURE

Peeche kaam kaise hota hai

Phone ring hone se lekar transcript score hone tak, har call ke peeche ka technical pipeline.

1. **Call ingress.** Twilio inbound PSTN call ko ek webhook pe route karta hai. Backend ek `Call` record kholta hai (Twilio call SID, from/to number, status = in_progress) aur conversation session shuru karta hai.
2. **Turn loop.** Har turn pe caller ki speech text mein convert hoti hai (ASR), transcript mein add hoti hai, aur LLM ko active agent ke persona/department prompt ke saath, 12 registered tools ki function-calling schema bhej di jaati hai. Model ya seedha jawab deta hai ya tool call karta hai; jawab us agent ki configured text-to-speech (TTS) voice mein bol ke wapas bheja jaata hai.
3. **RAG se grounding.** Coverage/policy wale sawaalon ke liye model `search_documents` call karta hai, jo Agent Studio mein upload kiye PDF/DOCX ke chunks pe vector-embedding similarity search chalata hai aur cited passages (filename, page, confidence) laata hai — bina source ke jawab nahi deta.
4. **Live data access.** Member-specific facts ke liye — eligibility, claim status, formulary tier/copay, in-network pharmacies — model typed tool handlers call karta hai jo seedha PBM data layer se query karte hain, wahi source of truth jo ek human CSR bhi dekhta.
5. **Per-turn confidence.** Har assistant message ko 0–100% score kiya jaata hai aur real-time mein Live Operations pe stream hota hai, taaki supervisor caller se pehle hi dekh le ki AI kahin unsure to nahi hai.
6. **Guardrails.** Model khud `escalate_to_human`, `schedule_callback`, ya `create_ticket` call kar sakta hai. Supervisor bhi Live Operations se kabhi bhi takeover force kar sakta hai — har turn ke start mein ek `ai_paused` flag check hota hai, isliye AI chup ho jaata hai aur call drop nahi hoti.
7. **Post-call processing.** Call khatam hote hi background jobs ek structured summary banate hain, sentiment classify karte hain, `resolution_status` set karte hain (resolved / escalated / abandoned), aur ek LLM-as-judge QA pass chalta hai jo accuracy, compliance, empathy, aur resolution — har ek 0–100 pe, written notes ke saath — score karta hai.
8. **Storage aur multi-tenancy.** Postgres calls, per-turn messages, tool-call traces, customer profiles, aur appointments — sab tenant ke hisaab se scoped — store karta hai. Angular frontend JWT se authenticate karta hai aur FastAPI REST layer se read/write karta hai.

RAG (Retrieval-Augmented Generation) — LLM ke jawab ko retrieve kiye gaye document chunks pe ground karna, model ki apni memory ke bharose nahi.

ASR / TTS — speech-to-text (awaaz→text) aur text-to-speech (text→awaaz).

Function / tool calling — LLM ka beech conversation mein hi ek typed backend action call kar dena, sirf text generate karne ki jagah.

LLM-as-judge — ek alag model pass use karke transcript ko ek rubric ke against score karna, jaise QA analyst karta hai.

Embeddings — text ke numeric vector representations, semantic similarity search ke liye.

Webhook — ek HTTP callback jo Twilio call events pe backend ko control handover karta hai.

03 — FEATURE WALKTHROUGH

Har screen: use case, tareeka, value, controls

Neeche har card ek page cover karta hai — us CSR scenario ke saath jiske liye woh bana, technical tareeka, business ke liye value, aur screen ka har button.

Dashboard — AI Workforce Overview

/dashboard

USE CASE

Ek contact-center manager har subah app kholta hai yeh dekhne ke liye ki pichhle 14 din mein AI workforce ne kaisa perform kiya — kitni calls aayin, kitne issues bina human ke resolve hue, aur usse kitna dollar bacha.

KAISE KAAM KARTA HAI

Summary endpoint Postgres se call records aggregate karta hai: status ke hisaab se counts, completed calls ke % ke roop mein resolution/escalation rate, rolling 14-din ka daily call-volume, aur latest 8 conversations agent/intent/sentiment/resolution ke saath.

VALUE

Ek hi screen batati hai "AI workforce theek chal raha hai ya nahi, aur paisa bach raha hai ya nahi" — bina ek bhi individual call khole.

CONTROL	KYA KARTA HAI
Stat tiles	Active calls, AI agents running, Resolution rate, Cost saved, Total calls, Appointments booked, Human transfer rate, Avg handle time — read-only KPI numbers (formulas Section 5 mein).
Call-volume chart	14 din ka daily call count; kisi bar pe hover karke exact number dekh sakte ho.
Recent conversations	Latest 8 calls — from-number, agent, intent, sentiment, aur resolution badge ke saath.

Agent Studio — AI employees banao aur manage karo

/agents

USE CASE

Ek supervisor chahta hai ek dedicated AI CSR pharmacy-benefits queue ke liye — Hindi aur English dono mein baat kare, apna persona ho, aur apne khud ke plan documents ho jinse jawab grounded hon — default agent se alag jo main line uthaata hai.

KAISE KAAM KARTA HAI

Har "AI employee" ek `Agent` record hai — naam, persona prompt, mapped TTS voice, aur department. "general" seedha calls uthata hai aur detected intent ke hisaab se callers ko specialists ki taraf route karta hai. Har agent ka apna "Knowledge base" panel hota hai jahan drag-and-drop se PDF/DOCX upload karte ho; har upload chunk hokar, embed hokar index ho jaata hai taaki `search_documents` call ke dauraan usko cite kar sake.

VALUE

Alag-alag business lines ko apne documents aur tone pe trained AI CSR milta hai — model retrain karne ki bajaye sirf persona aur knowledge base badal do.

CONTROL	KYA KARTA HAI
Search	Naam ya persona text se agent grid filter karta hai.
Preview	Chuni hui TTS voice ko browser mein bajaata hai (approximate stand-in) — commit karne se pehle sun sakte ho.
Create	Diye gaye naam/voice/persona/department ke saath naya AI employee bana deta hai.
Duplicate	Ek existing agent ka poora config copy karke naye agent ka starting point bana deta hai.
Delete	Confirmation ke baad agent ko permanently hata deta hai.
Save changes	Naam, voice, persona, ya department ki edits ko save karta hai.
Make default	Is agent ko main inbound line ka answering agent bana deta hai.
Browse files / drag-drop	Knowledge-base documents upload karta hai; chunking/embedding ke dauraan per-file progress bar dikhata hai.
Delete (document)	Ek uploaded document aur uske indexed chunks ko knowledge base se hata deta hai.

Live Operations — supervisor ka cockpit

/live

USE CASE

Ek supervisor floor ko waise hi dekh raha hai jaise human team ko dekhta. 30 calls live hain, aur ek caller ka confidence score 62% pe gir gaya kyunki AI prior-authorization ke ek edge-case pe unsure hai — supervisor ko dekhna hai kya ho raha hai aur bina call drop kiye step in karna hai.

KAISE KAAM KARTA HAI

Har 4 second mein in-progress calls poll hoti hain, live transcript preview aur latest per-turn confidence score ke saath, color-banded (good ≥90%, warning 70–89%, critical <70%). "Stop AI" call ka `ai_paused` flag on kar deta hai, jo har AI response se pehle check hota hai — AI chup ho jaata hai aur human ko takeover karna hota hai, bina underlying Twilio call ko drop kiye. "Send suggestion" ek coaching note likhta hai jo AI apne next turn pe padhta hai — bilkul whisper-coach jaisa, bina AI ko poora offline kiye.

VALUE

Yeh wahi safety net hai jo autonomous calling ko ek regulated, health-adjacent domain mein acceptable banata hai — kuch bhi fully unsupervised nahi chalta, aur human seconds mein intervene kar sakta hai.

CONTROL	KYA KARTA HAI
Monitor	Us call ke liye live call-detail drawer kholta hai — transcript, tool calls, aur ab-tak ka QA.
Stop AI / Resume AI	Us specific call ke liye human-takeover mode on/off karta hai.
Send suggestion	AI ko mid-call real-time coaching prompt bhejne ke liye inline box kholta hai.

Conversations — Calls & Appointments

/calls

USE CASE

Ek QA analyst ko dekhna hai ki ek call escalate kyun hui, ya compliance reviewer ko exact transcript chahiye ki AI ne member ko uski drug coverage ke baare mein kya bataya tha.

KAISE KAAM KARTA HAI

Har call from/to number, handling agent, aur detected intent ke saath list mein dikhti hai. Ek call kholne pe status, resolution, sentiment, AI confidence, LLM-generated call summary, four-axis QA score (accuracy / compliance / empathy / resolution, har ek independently judge hui written notes ke saath), aur poora turn-by-turn transcript tool-call traces aur RAG citations ke saath interleaved dikhta hai.

VALUE

Har AI decision exact data tak traceable hai jo usne use kiya — wahi audit trail jo health-plan compliance team pehli baar kisi jawab pe sawaal uthane pe maangegi.

CONTROL	KYA KARTA HAI
Status / Agent / Date filters	Call list ko narrow karte hain.
Row click	Us call ke liye call-detail drawer kholta hai.
Download .txt NEW	Open call ko plain-text file mein export karta hai: metadata, ek dedicated sentiment-analysis section (overall sentiment, AI confidence, empathy score), summary, QA scores, aur poora transcript — offline review ya audit trail ke liye.
Live Calls / Appointments tabs	Call log aur booked appointments ke beech switch karta hai.

Customers — AI ko kya yaad rehta hai

/customers

USE CASE

Ek returning caller ko har baar apna naam ya preferred language dobara batana nahi padna chahiye.

KAISE KAAM KARTA HAI

Customer profiles phone number se match hoti hain aur call ke dauraan `update_customer` tool se enrich hoti hain (naam, language), saath hi ek rolling `last_interaction` timestamp bhi update hota hai; profiles Data Import se bulk-load bhi ho sakti hain.

VALUE

Calls ke beech continuity — caller ko "yaad rakha gaya" feel hoga, "har baar naye sire se poochtaachh" nahi.

CONTROL	KYA KARTA HAI
Search	Naam ya phone number se filter karta hai.
Row click	Customer detail drawer kholta hai — profile aur interaction history.

Workflows — AI ke liye fixed procedures

/workflows

USE CASE

"Jab bhi caller apni pharmacy badalna chahe, pehle member verify karo, phir nayi pharmacy dhoondho, phir change confirm karo out loud" — ek scripted procedure jo AI ko exactly follow karna hai, improvise nahi.

KAISE KAAM KARTA HAI

Ek workflow ka plain-language trigger description hota hai (caller intent se match hota hai), optional department scope, aur steps ki ek ordered list — har step 12 registered tools mein se ek naam leta hai plus ek optional natural-language condition (jaise "only if member is verified"). Ek active, matching workflow us call ke liye AI ka checklist ban jaata hai.

VALUE

"LLM khud decide karta hai kya karna hai" aur "compliance ko har baar yehi exact sequence chahiye" — dono ke beech pull banata hai; jo hissa script hona chahiye, ho sakta hai.

CONTROL	KYA KARTA HAI
Create workflow	Naam, trigger phrase, aur department ke saath naya workflow define karta hai.
Activate / Deactivate	Delete kiye bina, live calls ke liye workflow ko on/off karta hai.
Delete	Confirmation ke baad workflow ko permanently hata deta hai.
+ Add step	Workflow ki sequence mein ek tool call add karta hai.
↑ / ↓	Kisi step ko sequence mein aage/peeche move karta hai.
Remove	Ek step ko delete karta hai.
Save changes	Workflow ka naam, trigger, department, active state, aur steps save karta hai.

Analytics — poore workspace ke trends

/analytics

USE CASE

Ek team lead jaanna chahta hai ki is hafte escalations badh rahe hain ya nahi, ya agli shift staff karne se pehle kaunse caller intents call volume pe hawi hain.

KAISE KAAM KARTA HAI

Wahi 14-din ki window ko top intents, sentiment mix, daily resolved/escalated/abandoned trend, aur daily positive/neutral/negative/frustrated sentiment trend mein aggregate karta hai.

VALUE

Individual call records ko ek pattern mein badal deta hai jis pe manager action le sake — jaise agar "prior authorization" intent spike kar raha hai to pharmacy specialist staff karo.

CONTROL	KYA KARTA HAI
Chart tooltips	Kisi point/bar pe hover karke uske peeche ka exact daily count dikhta hai.

AI Training Center — self-improvement loop

/training

USE CASE

AI ek nayi drug-tier change ke baare mein baar-baar shaky jawab de raha hai jo abhi uske knowledge base mein hai hi nahi — kisi ko yeh pattern pakadna hai isse pehle ki agli 10 calls mein bhi yehi ho.

KAISE KAAM KARTA HAI

"Run analysis" recent calls (low confidence scores, escalations, aur low QA scores waali) pe ek LLM pass chalata hai, unko insights mein cluster karta hai — missing_knowledge, prompt_improvement, ya workflow_improvement tag ke saath — har ek ke saath kitni supporting example calls hain, wo count bhi.

VALUE

"AI ne galat kiya" se seedha "yahan exactly kya upload/fix karna hai" tak loop close karta hai — supervisor ko sau transcripts manually padhne nahi padte.

CONTROL	KYA KARTA HAI
Run analysis	Recent calls ka fresh scan chalata hai improvement patterns ke liye.
All / New / Acknowledged / Dismissed	Insights ko unke review-status se filter karta hai.
Acknowledge	Insight ko "accepted / action ho raha hai" mark karta hai.
Dismiss	Insight ko "relevant nahi hai" mark karta hai.

Data Import — real operational data bulk-load karna

/data-import

USE CASE

Go-live se pehle team ko health plan ke systems se actual member roster, claims history, drug formulary, aur pharmacy network load karna hai — haath se type nahi karna.

KAISE KAAM KARTA HAI

CSV upload karo, batao yeh kya represent karta hai (members / claims / formulary / pharmacy), CSV columns ko expected fields se map karo (required wale marked hain), pehli 5 mapped rows preview karo, phir import karo. Wahi file dobara import karne pe existing records apne natural key se upsert ho jaate hain, duplicate nahi banate.

VALUE

AI workforce ko minutes mein real plan data pe ground kar deta hai — fixture data pe nahi — aur health plan ke apne extracts badalne pe usko sync bhi rakhta hai.

CONTROL	KYA KARTA HAI
What are you importing?	Target entity choose karta hai: members, claims, formulary, ya pharmacy.
CSV file picker	CSV load aur parse karta hai, header row padhta hai.
Column mapping	Har field ke liye ek dropdown, CSV header ko us field se map karta hai jo woh represent karta hai.
Import N row(s)	Import chalata hai aur har row ke created / updated / error count report karta hai.

04 — ACTION REGISTRY

AI call pe actually kya-kya kar sakta hai

Baarah tools, LLM ke apne judgment se mid-conversation callable (ya ek Workflow mein script kiye hue). Jo bhi call ek tool use karti hai, uska poora tool-call trace transcript mein log hota hai.

Tool	Kya karta hai	CSR moment jahan use hota hai
`search_documents`	Agent ke uploaded knowledge base pe RAG lookup; cited passages laata hai.	"90-day mail-order refill pe hamari policy kya hai?"
`verify_member`	Caller ki identity ko member records ke against confirm karta hai.	Har sensitive call ka shuruaat, PHI reveal karne se pehle.
`check_claim_status`	Submit hui claim ka adjudication status dekhta hai.	"Meri claim ka payment abhi tak kyun nahi hua?"
`get_benefits`	Plan ke benefit details retrieve karta hai.	"Is saal mera deductible kitna hai?"
`search_formulary`	Kisi drug ka tier, copay, aur prior-authorization requirement dekhta hai.	"Ozempic covered hai kya, aur prior auth chahiye?"
`find_pharmacy`	Kisi ZIP code ke paas in-network pharmacies dhoondhta hai.	"110001 ke paas kahan fill karwa sakta hoon?"
`schedule_appointment`	Appointment/consultation book karta hai.	"Kal 3pm pe kisi se callback ho sakta hai?"
`schedule_callback`	Bina full appointment ke ek simple callback queue karta hai.	Caller kaam mein busy hai, baad mein call chahta hai.
`escalate_to_human`	Call ko live-agent handoff ke liye flag karta hai aur resolution_status = escalated set karta hai.	AI resolve nahi kar paa raha, ya caller human maang raha hai.
`create_ticket`	Follow-up work ke liye ek support ticket kholta hai.	Issue ko call ke baad backend research chahiye.
`send_email`	Member ko confirmation/summary email bhejta hai.	"Yeh mujhe email kar sakte ho?"
`update_customer`	Seekhi hui profile facts (naam, language) wapas likh deta hai.	Pehli baar call karne wala apna naam batata hai.

05 — COST AUR RESOLUTION KA MATH

Dashboard ke tiles ke peeche ke numbers

"Resolution rate," "Human transfer rate," "Avg handle time," aur "Cost saved" exactly kaise calculate hote hain — round-figure marketing number nahi.

Resolution rate

```
resolution_rate = resolved_calls ÷ completed_calls × 100
```

Jitni calls poori hui, unme se kitna % AI ne akele close kar diya, bina kisi human ke involve hue. Yeh platform ka headline "first-call-resolution (FCR)" jaisa number hai.

Human transfer rate (escalation rate)

```
escalation_rate = escalated_calls ÷ completed_calls × 100
```

Resolution rate ka counterpart metric hai — kitni baar AI ne sahi se pehchana ki usko handoff karna chahiye (`escalate_to_human` ke through), jawab force karne ki bajaye.

Average handle time (AHT)

```
avg_handle_time = mean(ended_at − started_at) — completed calls pe
```

Wahi operational metric jo contact center pehle se human agents ke liye track karta hai — isliye AI aur human performance ko seedha compare kiya ja sakta hai.

Cost saved (estimate)

```
cost_saved_estimate = resolved_calls × $12.00
```

$12 ek stated benchmark assumption hai ek fully-loaded human-handled contact-center call ke liye — yeh backend ka ek constant hai, measurement nahi, aur dashboard isko estimate hi bataata hai, exact accounting nahi.

Beta note: $12/call figure ko placeholder samjho. Jab finance apna blended cost-per-contact de, usko yahan swap kar dena — har dashboard ka dollar figure automatically recalculate ho jaayega, koi aur change nahi karna padega.

QA score vs. resolution status — do alag judgments

`resolution_status` (resolved / escalated / abandoned) procedurally set hota hai — call mein actually kya hua, human involve hua ya nahi. Chaar QA scores (accuracy, compliance, empathy, resolution — har ek 0–100) ek independent LLM-as-judge review hai poore transcript aur tool-call trace ka. Ek call "resolved" ho sakti hai lekin curt tone ki wajah se empathy pe kat sakti hai, ya "escalated" sahi tarike se ho aur phir bhi compliance pe accha score kare — jaise member verification se pehle data disclose karne se sahi tarah se mana karna.

06 — DATA HANDLING

Security aur access, jaisa abhi bana hai

- Design se hi multi-tenant hai — har call, agent, customer, aur document ek `tenant_id` se scoped hai.
- Frontend aur API access JWT-authenticated hai; conversation data pe anonymous access nahi hai.
- AI jo bhi tool call ya message karta hai, transcript mein likha jaata hai, isliye har AI decision — aur jis data se usne wo decision liya — baad mein review ho sakta hai.
- Supervisor overrides (Stop AI, Send suggestion) us call ke against log hote hain jispe issue kiye gaye.

Beta note: is build ne abhi tak koi formal security ya regulatory compliance certification (jaise HIPAA) nahi liya hai. Isko ek pilot/evaluation environment samjho, production PHI handle karne ke liye certified system nahi — jab tak wo review complete na ho.

Yeh ek Beta release hai

Is document mein jo bhi likha hai, wo current build ka real, working behaviour hai — yahan kuch bhi aspirational nahi hai. Lekin platform abhi bhi young hai, aur pilot usage se jo nikalega usi ke hisaab se change hota rahega.

07 — JAANI-SAMJHI LIMITATIONS

- **Voice preview approximate hai.** Agent Studio ka "Preview" button chuni hui voice ko browser ke built-in text-to-speech se bajaata hai — production call voice kaisi sunayi degi, uska sirf stand-in hai.
- **Cost-saved ek placeholder estimate hai.** Poore workspace mein ek hi flat $12/call assumption use hoti hai, har tenant/call-type ke liye alag benchmark nahi.
- **Recent-conversations list abhi click-through nahi hai** Dashboard pe — poora call-detail drawer dekhne ke liye Conversations page use karo.
- **Abhi koi formal compliance certification nahi hai** — Section 6 dekho.

AAGE KYA

- Flat cost-saved assumption ko ek configurable, per-tenant cost-per-contact benchmark se replace karna.
- Deeper analytics drill-down (kisi trend point ko click karke uske underlying calls tak jaana).
- Workflow steps ke liye richer conditions aur branching.
- Kisi bhi production PHI handling se pehle formal security aur compliance review.

Is pilot se milne wala feedback seedha decide karega ki aage kya banega. Beta try karne ke liye shukriya — yeh document, aur product, dono improve hote rahenge.